

Unit-III

Functional Dependency

A functional dependency occurs when one attribute uniquely determines another attribute within a relation. It is a constraint that describes how attributes in a table relate to each other. If attribute A functionally determines attribute B we write this as the $A \rightarrow B$.

Functional dependencies are used to mathematically express relations among database entities and are very important to understanding advanced concepts in Relational Database Systems.

Example:

roll_no	name	dept_name	dept_building
42	abc	CO	A4
43	pqr	IT	A3
44	xyz	CO	A4
45	xyz	IT	A3
46	mno	EC	B2
47	jkl	ME	B2

From the above table we can conclude some valid functional dependencies:

- $\text{roll_no} \rightarrow \{ \text{name}, \text{dept_name}, \text{dept_building} \}$ → Here, roll_no can determine values of fields name, dept_name and dept_building, hence a valid Functional dependency

- $\text{roll_no} \rightarrow \text{dept_name}$, Since, roll_no can determine whole set of $\{\text{name}, \text{dept_name}, \text{dept_building}\}$, it can determine its subset dept_name also.
- $\text{dept_name} \rightarrow \text{dept_building}$, Dept_name can identify the dept_building accurately, since departments with different dept_name will also have a different dept_building
- More valid functional dependencies: $\text{roll_no} \rightarrow \text{name}$, $\{\text{roll_no}, \text{name}\} \twoheadrightarrow \{\text{dept_name}, \text{dept_building}\}$, etc.

Here are some invalid functional dependencies:

- $\text{name} \rightarrow \text{dept_name}$ Students with the same name can have different dept_name , hence this is not a valid functional dependency.
- $\text{dept_building} \rightarrow \text{dept_name}$ There can be multiple departments in the same building. Example, in the above table departments ME and EC are in the same building B2, hence $\text{dept_building} \rightarrow \text{dept_name}$ is an invalid functional dependency.
- More invalid functional dependencies: $\text{name} \rightarrow \text{roll_no}$, $\{\text{name}, \text{dept_name}\} \rightarrow \text{roll_no}$, $\text{dept_building} \rightarrow \text{roll_no}$, etc.

Types of Functional Dependencies in DBMS

1. Trivial functional dependency
2. Non-Trivial functional dependency
3. Multivalued functional dependency
4. Transitive functional dependency

1. Trivial Functional Dependency

In Trivial Functional Dependency, a dependent is always a subset of the determinant. i.e. If $X \rightarrow Y$ and Y is the subset of X , then it is called trivial functional dependency.

Symbolically: $A \rightarrow B$ is trivial functional dependency if B is a subset of A .

The following dependencies are also trivial: $A \rightarrow A$ & $B \rightarrow B$

Example 1 :

- $ABC \rightarrow AB$
- $ABC \rightarrow A$
- $ABC \rightarrow ABC$

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here, {roll_no, name} $\rightarrow$ name is a trivial functional dependency, since the dependent name is a subset of determinant set {roll_no, name}. Similarly, roll_no $\rightarrow$ roll_no is also an example of trivial functional dependency.

2. Non-trivial Functional Dependency

In **Non-trivial functional dependency**, the dependent is strictly not a subset of the determinant. i.e. If $X \rightarrow Y$ and **Y is not a subset of X**, then it is called Non-trivial functional dependency.

Example 1 :

- Id $\rightarrow$ Name
- Name $\rightarrow$ DOB

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18

roll_no	name	age
44	xyz	18

Here, $\text{roll_no} \rightarrow \text{name}$ is a non-trivial functional dependency, since the dependent name is not a subset of determinant roll_no. Similarly, $\{\text{roll_no}, \text{name}\} \rightarrow \text{age}$ is also a non-trivial functional dependency, since age is not a subset of $\{\text{roll_no}, \text{name}\}$

3. Semi Non Trivial Functional Dependencies

A semi non-trivial functional dependency occurs when part of the dependent attribute (right-hand side) is included in the determinant (left-hand side), but not all of it. This is a middle ground between trivial and non-trivial functional dependencies. $X \rightarrow Y$ is called semi non-trivial when $X \cap Y$ is not NULL.

Example:

Consider the following table:

Student_ID	Course_ID	Course_Name
101	CSE101	Computer Science
102	CSE102	Data Structures
103	CSE101	Computer Science

Functional Dependency:

$\{\text{StudentID}, \text{CourseID}\} \rightarrow \text{CourseID}$

This is semi non-trivial because:

- Part of the dependent attribute (Course_ID) is already included in the determinant ($\{\text{Student_ID}, \text{Course_ID}\}$).
- However, the dependency is not completely trivial because $\{\text{StudentID}\} \rightarrow \text{CourseID}$ is not implied directly.

4. Multivalued Functional Dependency

In Multivalued functional dependency, entities of the dependent set are not dependent on each other. i.e. If $a \twoheadrightarrow \{b, c\}$ and there exists no functional dependency between b and c , then it is called a multivalued functional dependency.

Example:

bike_model	manuf_year	color
tu1001	2007	Black
tu1001	2007	Red
tu2012	2008	Black
tu2012	2008	Red
tu2222	2009	Black
tu2222	2009	Red

In this table:

- **X:** bike_model
- **Y:** color
- **Z:** manuf_year

For each bike model (bike_model):

1. There is a group of colors (color) and a group of manufacturing years (manuf_year).

2. The colors do not depend on the manufacturing year, and the manufacturing year does not depend on the colors. They are independent.
3. The sets of color and manuf_year are linked only to bike_model.

That's what makes it a multivalued dependency.

In this case these two columns are said to be multivalued dependent on bike_model. These dependencies can be represented like this:

5. Transitive Functional Dependency

In transitive functional dependency, dependent is indirectly dependent on determinant. i.e. If $a \rightarrow b$ & $b \rightarrow c$, then according to axiom of transitivity, $a \rightarrow c$. This is a transitive functional dependency.

Example:

enrol_no	name	dept	building_no
42	abc	CO	4
43	pqr	EC	2
44	xyz	IT	1
45	abc	EC	2

Here, $\text{enrol_no} \rightarrow \text{dept}$ and $\text{dept} \rightarrow \text{building_no}$. Hence, according to the axiom of transitivity, $\text{enrol_no} \rightarrow \text{building_no}$ is a valid functional dependency. This is an indirect functional dependency, hence called Transitive functional dependency.

6. Fully Functional Dependency

In full functional dependency an attribute or a set of attributes uniquely determines another attribute or set of attributes. If a relation R has attributes X, Y, Z with the dependencies $X \rightarrow Y$ and $X \rightarrow Z$ which states that those dependencies are fully functional.

7. Partial Functional Dependency

In partial functional dependency a non key attribute depends on a part of the composite key, rather than the whole key. If a relation R has attributes X, Y, Z where X and Y are the composite key and Z is non key attribute. Then $X \rightarrow Z$ is a partial functional dependency in RBDMS.

